

Case Study

Smart Energy Metering Solution by SLS for Medium Scale Establishments like Military Cantonments

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Typographic Conventions

This document uses the typographic conventions as shown below:

Visual Cue	Meaning
Bold Type with Initial Capital letters	All headings and Sub headings Titles in a document are displayed in bold type with initial capital letters; Example: General Description, Features.
Bold Type with Italic Letters	All Definitions, Figure and Table Headings are displayed in Italics.
1., 2.	Numbered steps are used in a list of items, when the sequence of items is important. such as steps listed in procedure.
•	Bullets are used in a list of items when the sequence of items is not important.

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1 Introduction

SLS, a leading technology provider, has successfully implemented a comprehensive Smart Energy Metering solution across several military cantonments in India. These establishments required an efficient, scalable, and unified system capable of managing a wide range of energy metering operations. SLS delivered a fully integrated solution that addressed these needs while enhancing energy management, optimizing operations, and improving consumer experience.

2 Key Challenges faced by Military Cantonments

Before the deployment of SLS's solution, military cantonments faced several challenges in managing energy consumption and billing:

1. **Manual Meter Readings:** Energy meters were read manually, leading to inefficiencies and errors.
2. **Inefficient Billing System:** Existing billing systems did not support prepaid billing or integration with consumer wallets.
3. **Diverse Meter Types:** The cantonments utilized meters from different manufacturers, making integration difficult.
4. **Communication Complexity:** Meters with different communication protocols (GPRS and RF) required a unified system to handle all data transmission.
5. **Consumer Engagement:** Lack of a user-friendly consumer portal and mobile app for energy consumption tracking and recharging prepaid accounts.

3 The SLS Solution

SLS developed a smart and efficient energy metering system designed to address all these challenges. The solution includes several critical modules and features to provide seamless management of energy consumption and billing for the military cantonments:

1. Unified Meter Reading and Data Collection

SLS's system supports the integration of smart meters from various manufacturers that comply with the IS16444 (DLMS) standard. It can read meter data using different communication interfaces, including GPRS and RF. This ensures flexibility in managing meters from multiple suppliers. The system can scale to read data from a large number of meters, using both push and pull options for data collection.

2. Data Validation and Storage

The system ensures that data read from meters is validated for accuracy before being stored in a secure, centralized database. This data is crucial for accurate billing and operational insights.

3. Prepaid Billing and Wallet Management

The prepaid billing engine processes meter data at predefined intervals to determine energy consumption and calculate charges based on the billing profile assigned to each Consumer. The system automatically deducts the corresponding amount from the consumer's wallet. Consumers receive notifications via email or SMS when their wallet balance falls below a specified threshold. If the balance becomes negative, the system automatically disconnects the meter's relay, cutting off power to the consumer's premises.

However, to ensure consumer convenience, the system allows for ****Friendly Hours****, during which power disconnection is delayed even if the balance is negative.

4. Consumer Engagement through Portal and Mobile App

Consumers can track their energy consumption, view billing information, and recharge their prepaid accounts through a user-friendly consumer portal and mobile app. The solution supports various payment methods, including net banking, credit/debit cards, and UPI. If a consumer's power is disconnected due to a negative balance, recharging the account instantly triggers the system to send a ****Relay Connect**** command, restoring power at the consumer's premises.

5. Asset Management

The system includes an asset management module that tracks all meter-related operations. It manages meter associations with consumers, records meter replacements, and logs the history of each meter's operational lifecycle. This module ensures a clear and complete overview of meter management, which is vital for efficient operations.

6. Consumer and Billing Management

The solution's consumer management and prepaid billing modules cover all consumer-related activities, including the creation of new consumers, assigning meters, setting billing profiles, and managing energy charges. Billing profiles can be customized for each consumer, including energy charges, fixed charges, and recharge/refund histories. The system also generates detailed billing reports, which provide transparency and insight into consumption patterns.

4 Benefits of the SLS Solution

The deployment of SLS's Smart Energy Metering system at military cantonments has led to several key benefits: - Operational Efficiency: Automated meter readings, data validation, and billing have reduced manual intervention and human error. - Scalability: The system's ability to handle a large number of meters and different communication interfaces has ensured seamless operations, even as the cantonments expand their energy infrastructure. - Prepaid Billing Optimization: The prepaid billing engine and wallet integration have improved revenue collection and empowered consumers to manage their energy usage proactively. - Consumer Empowerment: The consumer portal and mobile app have made energy management easier for users, allowing them to monitor consumption and recharge accounts in real-time. - System Reliability: The system's friendly hours feature and automated relay commands ensure uninterrupted power supply during critical periods, enhancing user satisfaction.

5 Conclusion

SLS's unified Smart Energy Metering solution has revolutionized the way military cantonments in India manage their energy consumption and billing processes. The system's flexibility, scalability, and comprehensive features have streamlined operations, reduced inefficiencies, and improved the overall consumer experience. By integrating all aspects of energy metering, asset management, consumer engagement, and billing into a single system, SLS has set a new standard for smart energy solutions in similar large-scale establishments.